

Supplemental Material: Auditing Trajectory-Based Transfer Diagnostics in Variational Quantum Learning

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This Supplemental Material collects the supporting analyses referenced in the main text: the concentration of the identity prefactor (Sec. S1), the one-step versus multi-step identity check (Sec. S2), the association of alignment with the query loss rather than the gap (Sec. S3), the near-zero-mean-transfer character of the task family (Sec. S4), and four robustness checks—a term-type-stratified split, additional Hamiltonian families and entangler topologies, a finite-shot variance proxy, and a random initialization (Sec. S5); and a synthetic implementation check showing that the joint adjustment does not mechanically zero an explicitly injected direct A component (Sec. S6). All quantities use the definitions, estimators, and hierarchical-bootstrap procedure of the main text. Rank correlations are Spearman coefficients; joint controls are partial rank correlations $\rho(A, G | I_{\text{val}}, L_{\text{val}}(\theta_0))$ unless stated otherwise. The underlying per-task tables are provided in the accompanying data repository.

S1. CONCENTRATION OF THE IDENTITY PREFACTOR

The first-order identity of the main text reads $I_{\text{val}} \approx \alpha A q$ with prefactor $q = \|g_{\text{tr}}\| \|g_{\text{val}}\|$. The small-relative-variation limit of q is a statement about the relative spread of q , not of $\|g_{\text{tr}}\|$ alone. Table SI reports the pooled coefficient of variation of $\|g_{\text{tr}}\|$, $\|g_{\text{val}}\|$, and q across tasks at each system size. Both gradient norms concentrate with scale, and the product q —being the product of two fluctuating norms—has a larger but likewise decreasing coefficient of variation. This is the regime in which the ranking of I_{val} approaches, in the small-relative-variation limit, that of A .

TABLE SI. Pooled coefficient of variation of the two gradient norms and of the identity prefactor $q = \|g_{\text{tr}}\| \|g_{\text{val}}\|$, as a function of system size n (pooled over depth L and seeds).

n	$\text{CV}(\ g_{\text{tr}}\)$	$\text{CV}(\ g_{\text{val}}\)$	$\text{CV}(q)$
4	0.336	0.361	0.551
6	0.285	0.282	0.466
8	0.246	0.261	0.406
10	0.264	0.261	0.451

S2. ONE-STEP VERSUS MULTI-STEP IDENTITY

The identity is a first-order (tangent) expression; it is at its most accurate in the one-step limit. Table SII compares the rank correlation between the alignment A and the validation improvement measured after a single inner step ($I_{\text{val}}^{(1)}$) with that measured after the full $K = 10$ inner steps ($I_{\text{val}}^{(K)}$). At $K = 1$ the correlation is near-perfect and essentially flat in n (0.95–0.97), confirming that the tangent expression captures the one-step improvement directly. At $K = 10$ the correlation is lower and rises with scale (0.60 to 0.85): as gradient magnitudes and adaptation steps shrink while norm variability also changes, the full trajectory moves closer to its initial tangent, and the identity increasingly predicts the multi-step improvement as well. The tightening reported in the main text is therefore the convergence of the K -step improvement toward the one-step tangent, not a change in the identity itself.

S3. ALIGNMENT VERSUS THE QUERY LOSS

The main text controls the apparent association between A and the observable-generalization gap G . The same conclusion holds when the target is the query loss $L_{\text{val}}(\theta_K)$ itself rather than the gap. Table SIII reports $\rho(A, L_{\text{val}}(\theta_K))$ raw, controlling for the local improvement I_{val} , and controlling jointly for I_{val} and the initial baseline $L_{\text{val}}(\theta_0)$. The

TABLE SII. Rank correlation of the support-validation alignment A with the validation improvement after one inner step ($I_{\text{val}}^{(1)}$) and after $K = 10$ inner steps ($I_{\text{val}}^{(K)}$), pooled over depth and seeds.

n	$\rho(A, I_{\text{val}}^{(1)})$	$\rho(A, I_{\text{val}}^{(K)})$
4	+0.95	+0.60
6	+0.96	+0.68
8	+0.97	+0.76
10	+0.97	+0.85

raw association is moderate and negative but is near zero under the joint control for each pooled n ($|\rho| \leq 0.10$), so the vanishing residual is not special to the signed gap.

TABLE SIII. Alignment versus the query loss $L_{\text{val}}(\theta_K)$: raw rank correlation, controlling for the local improvement I_{val} , and the joint control additionally removing the initial baseline $L_{\text{val}}(\theta_0)$.

n	$\rho(A, L_{\text{val}}^K)$	$\rho(A, L_{\text{val}}^K I_{\text{val}})$	$\rho(A, L_{\text{val}}^K I_{\text{val}}, L_{\text{val}}^0)$
4	-0.56	-0.34	-0.06
6	-0.49	-0.39	-0.03
8	-0.45	-0.38	-0.10
10	-0.36	-0.38	-0.02

We also report the baseline-corrected gap $\Delta G = [L_{\text{val}}(\theta_K) - L_{\text{tr}}(\theta_K)] - [L_{\text{val}}(\theta_0) - L_{\text{tr}}(\theta_0)] = I_{\text{tr}} - I_{\text{val}}$, which removes the initial baseline at the level of the target. Its raw association with A is $\rho(A, \Delta G) = -0.48, -0.55, -0.53, -0.59$ at $n = 4, 6, 8, 10$; controlling for I_{val} alone removes it, $\rho(A, \Delta G | I_{\text{val}}) = +0.03, -0.03, -0.07, -0.10$.

S4. NEAR-ZERO-MEAN-TRANSFER REGIME

Table SIV quantifies the transfer content of the primary task family. Support adaptation improves the support loss substantially (mean I_{tr} from +0.41 to +0.16), but the mean validation improvement is statistically indistinguishable from zero at every size, with roughly half of tasks improving. The per-task spread of I_{val} remains nonzero and is the quantity the first-order identity predicts. This regime is empirically a near-zero-mean-transfer setting under the specified construction: coefficients are drawn independently and, for the linear observable loss over disjoint Pauli support and query subsets, no shared minimizer is deliberately imposed; the family empirically exhibits little mean observable transfer to detect.

TABLE SIV. Transfer content of the primary task family: mean and standard deviation of the validation improvement I_{val} , fraction of tasks with $I_{\text{val}} > 0$, and mean support improvement I_{tr} .

n	$\langle I_{\text{val}} \rangle$	$\text{std}(I_{\text{val}})$	$\text{frac}(I_{\text{val}} > 0)$	$\langle I_{\text{tr}} \rangle$
4	-0.008	0.283	0.47	0.410
6	+0.002	0.161	0.51	0.313
8	-0.003	0.096	0.49	0.242
10	+0.001	0.060	0.50	0.158

S5. ROBUSTNESS CHECKS

The four checks below vary one ingredient of the setup at a time and confirm that the identity link and the near-zero joint-adjusted pooled gap residual both persist.

A. Term-type-stratified support/query split

To rule out that the gap is driven by an unbalanced split, we stratify the support/query split so that XX , YY , ZZ , and Z terms are balanced across the two subsets, and compare against the uniform (random) split. Table SV shows that both splits preserve the high identity link and the near-zero joint-controlled gap association.

TABLE SV. Uniform versus term-type-stratified support/query split. The identity link $\rho(A, I_{\text{val}})$ and the joint-controlled gap association $\rho(A, G | I_{\text{val}}, L_{\text{val}}(\theta_0))$ are preserved under stratification.

n	split	$\rho(A, I_{\text{val}})$	$\rho(A, G I_{\text{val}}, L_{\text{val}}^0)$
4	uniform	0.35	−0.29
4	stratified	0.50	+0.10
6	uniform	0.81	−0.03
6	stratified	0.72	−0.05
8	uniform	0.76	+0.04
8	stratified	0.86	−0.01

B. Additional Hamiltonian families and topologies

We repeat the analysis at $n = 6$ for three Hamiltonian families—the anisotropic XYZ family of the main text, a transverse-field Ising model (TFIM), and a fixed random-local-Pauli family—each with a linear-chain and a ring entangler. Table SVI shows that the identity link stays high ($\rho(A, I_{\text{val}}) = 0.49$ – 0.91) and the joint-controlled gap association stays near zero (-0.17 to $+0.13$) throughout, so the observed identity-link and adjusted-null association pattern is not specific to the XYZ family or the linear chain.

TABLE SVI. Identity link and joint-controlled gap association across three Hamiltonian families and two entangler topologies ($n = 6$, $L = 3$).

family	topology	$\rho(A, I_{\text{val}})$	$\rho(A, G I_{\text{val}}, L_{\text{val}}^0)$
XYZ	linear	0.81	−0.03
XYZ	ring	0.89	+0.08
TFIM	linear	0.49	+0.13
TFIM	ring	0.77	+0.07
rand. Pauli	linear	0.78	−0.03
rand. Pauli	ring	0.91	−0.17

C. Finite-shot variance proxy

The observable-variance proxy V_{tr} is defined with a finite shot budget. To confirm that finite sampling does not remove the effect, we compare V_{tr} estimated at 100 and 1000 shots against the exact ($N_{\text{shot}} \rightarrow \infty$) value across tasks. The rank agreement is $\rho = +0.84$ at 100 shots and $\rho = +0.98$ at 1000 shots, with a small positive sampling bias (mean offset $+0.003$ and $+0.001$, respectively). Finite-shot statistics therefore add, rather than remove, sources of the apparent association.

D. Random initialization

Finally, we replace the Reptile meta-initialization with a plain random initialization ($\theta \sim \mathcal{N}(0, 0.1^2)$, no meta-training) at $n = 6$, $L = 3$. The identity link is $\rho(A, I_{\text{val}}) = +0.74$, the raw gap association is $\rho(A, G) = -0.27$, and it collapses under the joint control to $+0.07$ —matching the Reptile arm ($0.68, -0.42, -0.01$). The identity and the

adjusted-null pattern are thus observed under random initialization as well, indicating that they are not specific to the meta-initialization; this comparison does not identify a unique causal mechanism.

S6. SYNTHETIC IMPLEMENTATION CHECK: INJECTED COMPONENTS ARE NOT MECHANICALLY ZEROED

Every demonstration in the main text is a collapse case, so one may ask whether the joint control is simply too aggressive—whether it would erase an explicitly injected alignment–gap component. A closed-form synthetic ensemble answers this. Per task we draw an alignment A , a norm product q with CV ≈ 0.45 (matching the experiments), the identity-channel improvement $I_{\text{val}} = \alpha A q + \varepsilon$, and an independent baseline $L_{\text{val}}(\theta_0)$; the gap is then constructed in two ways. In the *channels-only* construction, $G = -0.8 I_{\text{val}} + 0.8 L_{\text{val}}(\theta_0) + \varepsilon'$, so any A – G association flows entirely through the two controlled channels. In the *direct-channel* construction a direct term $-0.5 A$ is added, representing a real alignment→gap effect beyond local improvement and baseline. Three seeds of 150 tasks each give Table SVII: the channels-only association collapses under the joint control ($-0.65 \rightarrow -0.06$), exactly as in the physical experiments, while the direct-channel association survives it ($-0.90 \rightarrow -0.62$; per-seed range $[-0.66, -0.58]$). The numerical values depend on the injected channel strength: sweeping the direct coefficient from 0.1 to 0.8 (channel noise s.d. 0.10) gives surviving joint-controlled correlations of -0.21 , -0.33 , -0.45 , -0.62 , and -0.77 , so survival is robust down to injected strengths comparable to the channel noise. The controls of the audit protocol therefore remove channel-mediated associations while not mechanically forcing an explicitly injected component to zero: a diagnostic that survives them remains a candidate requiring independent validation.

TABLE SVII. Synthetic implementation check. Rank correlation of alignment with the gap, raw and under the joint control, when the gap contains only the local-improvement and baseline channels versus when an explicitly injected direct alignment channel is added (3 seeds $\times$ 150 tasks, pooled).

construction	$\rho(A, G)$	$\rho(A, G \mid I_{\text{val}})$	$\rho(A, G \mid I_{\text{val}}, L_{\text{val}}^0)$
channels only	-0.65	-0.01	-0.06
direct channel	-0.90	-0.45	-0.62